

University of Groningen

Density-dependent individual variation in male attractiveness in a wild field cricket

Niemelä, Petri T.; Tiso, Stefano; Dingemanse, Niels J.

Published in:
Behavioral Ecology

DOI:
[10.1093/beheco/arab009](https://doi.org/10.1093/beheco/arab009)

IMPORTANT NOTE: You are advised to consult the publisher's version (publisher's PDF) if you wish to cite from it. Please check the document version below.

Document Version
Publisher's PDF, also known as Version of record

Publication date:
2021

[Link to publication in University of Groningen/UMCG research database](#)

Citation for published version (APA):

Niemelä, P. T., Tiso, S., & Dingemanse, N. J. (2021). Density-dependent individual variation in male attractiveness in a wild field cricket. *Behavioral Ecology*, 32(4), 707–716.
<https://doi.org/10.1093/beheco/arab009>

Copyright

Other than for strictly personal use, it is not permitted to download or to forward/distribute the text or part of it without the consent of the author(s) and/or copyright holder(s), unless the work is under an open content license (like Creative Commons).

The publication may also be distributed here under the terms of Article 25fa of the Dutch Copyright Act, indicated by the "Taverne" license. More information can be found on the University of Groningen website: <https://www.rug.nl/library/open-access/self-archiving-pure/taverne-amendment>.

Take-down policy

If you believe that this document breaches copyright please contact us providing details, and we will remove access to the work immediately and investigate your claim.

Downloaded from the University of Groningen/UMCG research database (Pure): <http://www.rug.nl/research/portal>. For technical reasons the number of authors shown on this cover page is limited to 10 maximum.



Original Article

Density-dependent individual variation in male attractiveness in a wild field cricket

Petri T. Niemelä^{*}, Stefano Tiso, and Niels J. Dingemanse^{*}

Behavioural Ecology, Department of Biology, Ludwig-Maximilians University of Munich, Großhaderner Str. 2, Planegg-Martinsried, Germany

Received 30 July 2020; revised 22 January 2021; editorial decision 24 January 2021; accepted 26 January 2021; Advance Access publication 21 April 2021.

Social environments modify a male's ability to attract females and thus affect its fitness. Theory implies that an individual's fitness should trade-off with its ability to cope with competition. Individuals are expected to solve this trade-off differently: some males should be more attractive at low but others instead at high density. This prediction has rarely been tested in the wild. We used an automated RFID-surveillance system to quantify for each hour of the day, over 30 days (i.e., almost the entire adult lifespan of our model organism), whether a male had attracted a female in its burrow. The data were collected across a range of naturally varying local densities in wild field crickets, *Gryllus campestris*. We also estimated whether the shape of the relationship between attractiveness and density was under selection. At the population level, attractiveness increased from low to intermediate density, suggesting an Allee effect. Attractiveness subsequently declined at higher densities, for example, because of detrimental effects of increased competition. Opposite to expectations, males that were more attractive under low densities were also more attractive under higher densities. However, the increase in attractiveness with density varied among males, suggesting that Allee effects were individual-specific. Finally, selection was not acting on density-dependent attractiveness but males that lived longer acquired more mating partners. Our study reveals that social environments shape attractiveness in wild male insects, and imply the occurrence of individual-specific Allee effect that may be evolvable.

Key words: Allee effect, competition, individual variation, population density, reaction norm, social environment

INTRODUCTION

Males differ in their ability to attract mating partners and adopt different reproductive strategies (Andersson 1994; Gerhardt and Huber 2002). Moreover, costs and benefits associated with a given male reproductive strategy may vary as a function of the environment (Bateson and Healy 2005). For example, components of the social environment (e.g., population density) play a major role in shaping male's ability to attract females since they partly define the costs and benefits associated with expression of sexually selected traits in males (Bateson and Healy 2005). Indeed, the Allee effect theory posits that population level average mating success should, at low population densities, increase with increasing population density due to, for example, because mating partner are more easily found in larger groups (Courchamp et al. 1999; Stephens et al. 1999). Moreover, males vary in their ability to cope with intrasexual competition, and recent (verbal and formal) models suggests trade-offs between this ability and an individual's fitness (Lande et al. 2009; Engen et al. 2013; Wright et al. 2019), suggesting an important role for the social environment as agent of

selection. Specifically, variation in how such trade-offs are resolved should give rise to density-dependent alternative reproductive strategies (Lande et al. 2009; Engen et al. 2013; Wright et al. 2019). This leads to a testable prediction that individuals or genotypes with relatively high fitness under low densities perform relatively poorly under high densities, while individuals with high fitness under high densities perform relatively poorly under low densities (Lande et al. 2009; Engen et al. 2013; Wright et al. 2019).

Support for the prediction above comes from recent field studies on birds. These studies demonstrate that selection on phenotypic traits, such as clutch size (Sæther et al. 2016), aggression (Duckworth and Badyaev 2007), or exploratory behaviour (Nicolaus et al. 2016) fluctuate among years, from positive in low-density years to negative in high-density years. An implicit prediction of models incorporating density-dependent selection is that individuals or genotypes differ in how components of fitness vary as a function of density. This prediction has rarely been tested as most previous studies used cross-sectional (i.e., single measurement of fitness measured in one density for each individual) rather than longitudinal (i.e., repeated measurements of fitness measured in multiple densities for each individual) analyses (Duckworth and Badyaev 2007; Nicolaus et al. 2016). An exception is the

Address correspondence to P.T. Niemelä. E-mail: petri.t.niemela@gmail.com.

longitudinal study by Sæther et al. (2016) that demonstrated the existence of individual differences in how fitness varies as a function of population density in great tits (*Parus major*). The paucity of empirical demonstrations of individual variation in how fitness proxies vary as a function of density is not surprising given that most empirical studies focus on systems (e.g., birds) where relatively few individuals survive across time periods over which density varies most strongly (e.g., reproductive seasons, years). Here, we focus on an insect where local population density fluctuates *within* the lifetime of an individual because it varies spatiotemporally within a reproductive season, enabling the estimation of individual differences how a fitness-related trait varies as a function of local density.

We used the European field cricket (*Gryllus campestris*) as a highly suitable model. The European field cricket is a species with an annual life cycle, where males can acquire matings over the entire reproductive season (spanning from May through June), while experiencing temporal variation in local population density (Figure 1) (Rodríguez-Muñoz et al. 2011; Fisher, James, et al. 2015; Niemelä et al. 2015; Niemelä and Dingemanse 2017). Males attract females by acoustically signaling in front of a burrow; the area surrounding the burrow is also used for mating (Rodríguez-Muñoz et al. 2010; Rodríguez-Muñoz et al. 2011). These acoustic signals are also used to evaluate the structure of the local social environment and to compete with other males (Gerhardt and Huber 2002). In field crickets, local male density affects the intensity of competition among males for access to females (Gerhardt and Huber 2002). In this study, we therefore measured competition experienced by males by quantifying local male density. This allowed us to test whether the reaction norm describing a male's attractiveness as a function of local male density varied between individual males. We designed an automated detection system by fitting all burrows with a reader registering entering males and females fitted with a passive integrated transponder (PIT a.k.a RFID-tag). This setup allowed us to repeatedly measure each male's "attractiveness," defined here as the probability that a female was observed in the burrow area of a focal male within a given time frame. The transponder-reader system enabled us to quantify each male's attractiveness for each

hour of its adult life. Because of this, and because local density varied strongly within and among hours, days and burrows, we were able to estimate how attractiveness varied *within* individual males as a function of local density.

Here, we study whether males differ in how their attractiveness varies with local density of competitors, whether the data supported a predicted trade-off between attractiveness under high versus low densities and whether this variation is under selection. Based on the theory detailed above, we predicted that attractiveness varied as a function of density at the population level according to Allee effect theory (Courchamp et al. 1999; Stephens et al. 1999), but that the effect differed among individual males (either due to permanent environmental effects or additive genetic effects; not separated in our work). We expected that some males would be most attractive when competition was intense, that is, at high local density, while others would instead be most attractive when competition was relaxed, that is, at low local density. Finally, we assumed that attractiveness represented a fitness-related trait as attracting a female is a prerequisite for securing matings (Gerhardt and Huber 2002). We therefore predicted that the reaction norm characterizing how a male's attractiveness varied with local male density was under selection. We expected that selection would overall favor individuals that were attractive under low densities. This was because the year of study was a relatively low-density year (see Methods). A male's total number of unique female partners over the entire season predicts its reproductive fitness (Rodríguez-Muñoz et al. 2011). We therefore used this metric, that is, number of unique partners on the same burrow, as our proxy for reproductive fitness. Because we expected longer-lived males to attract a larger number of unique females over their adult lives, we further tested whether any selection on an individual's attractiveness-density reaction norm was mediated by a male's longevity.

METHODS

Study site and species

The data were collected in a 50 m × 50 m field plot near the Max Planck Institute for Ornithology (Seewiesen, southern Germany):

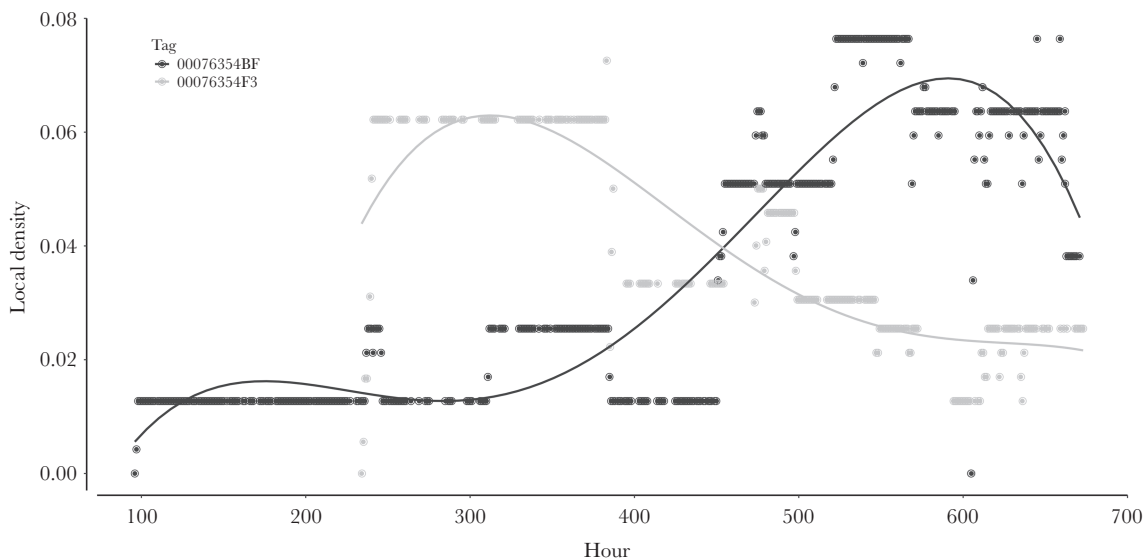


Figure 1

Variation in local density across unique, cumulative, hours throughout the season in two males (black line and circles, gray line and circles) within 5-m radius from the focal burrow. Lines indicate average density experienced by each male and the circles indicate data points for density for each focal hour.

47°58'35.5"N, 11°14'04.5"E) from 8 May to 6 June 2015. The study area was delimited by transparent plexiglass panels, high enough to prevent crickets from entering or leaving the area (Niemelä and Dingemanse 2017). Our enclosure nevertheless allowed ample movement because of its large relative size: common field crickets typically move only over short distances (on average 35 m) following post-wintering hibernation (Ritz and Köhler 2007). In early spring, all subjects emerged as adults within our study area from natural (self-made) burrows, where they had overwintered, and from which they could move freely. In this species, nymphs become active in early spring (March) and stay in close proximity of their burrow (c. 10–20 cm distance). After transforming into mature adults (the stage at which our data were collected), both sexes start searching for mating partners (Rodríguez-Muñoz et al. 2011; Niemelä et al. 2015). Given that crickets spend the majority of their time in, or nearby, burrows, they can easily be trapped, marked, and their behavior recorded for their entire adult life in great detail (Rodríguez-Muñoz et al. 2010, 2011; Fisher, David, et al. 2015; Niemelä et al. 2015; Niemelä and Dingemanse 2017).

Data collection

To mark individuals, they were trapped using custom-made traps that consisted of two parts: a 30-cm-long tube inserted into the entrance of the underground part of the burrow, and a cylinder-shaped chamber, in which the cricket would fall after climbing the tube while attempting to leave the burrow (Niemelä and Dingemanse 2017). Each field cricket was captured from its natal burrow 1–2 days after reaching sexual maturation (early May) ($n = 90$; 59 females, 31 males) and marked with a small circular numbered “bee” tag (diameter 2.5 mm), and fitted with unique RFID-tag (Passive Integrated Transponder: length 8 mm, diameter 1.4 mm; Trovan Ltd. Isle of Man). RFID-tags combined with RFID-antennae (see below) allow automated monitoring of each uniquely tagged cricket visiting each territory (burrow area) occupied with an antenna. Tags were attached to the pronotum using cyanoacrylate adhesive. Other studies have attached objects of similar size and weight on field crickets (Rodríguez-Muñoz et al. 2010, 2011; Fisher, David, et al. 2015). In addition, an experiment conducted in 2014 showed that RFID-tags do not affect survival in the wild over the period of 20 days for tagged versus untagged individuals ($n = 20$ individuals in each treatment): (daily survival probability (95% credible intervals) = 0.99 (0.96–1.00) versus 0.98 (0.95–1.00), respectively; Niemelä et al., unpublished data).

Our RFID-antennae read the identity of the RFID-tags with a date/time stamp each time a PIT-tagged cricket entered the reading range (1 data point/second when the cricket was in the antenna reading range; ~1–2 cm beyond both lateral sides of the antenna). The data were subsequently sent wirelessly to a computer in a nearby building (~500 m away) via main reader units (each controlling 20–25 antennae). We placed one circular RFID-antenna (Trovan Ltd. Isle of Man, diameter: 40 mm) at each burrow few days before the inhabitant was predicted to transform into an adult. We placed each antenna such that it covered the burrow entrance (Supplementary Figure S1). Crickets thus had to walk through the antenna to enter or exit the burrow. Antennae were positioned to not obstruct movement into and out of the burrow (Supplementary Figure S1). Crickets tend to stay in close proximity to the burrow entrance, since the burrow functions as a refuge against predators (Gawalek et al. 2014), and move in and out many times per hour. This aspect of their behavior results in high hourly detection

probabilities. We determined the X and Y coordinates of each burrow (Supplementary Figure S2) and noted the identity of each antenna installed at each burrow. Each antenna transmitted its identity when transmitting a reading, which consequently enabled us to collect data on the location of each individual in the area across the study period. We monitored a total number of 124 burrows using a total number of 94 antennae. The number of burrows exceeds the number of antennae because we removed antennae away from burrows that were abandoned or destroyed, and placed them at newly found burrows that crickets sometimes dug in response (34 burrows throughout the season).

We scored 21 (30-min long) videos recorded from 15 different burrows to calibrate our RFID-based readings. This showed that the RFID-antennae failed to detect 6.5% of cases where crickets moved into the antenna range. Given that each male was on average read 105.0 (SD = 73.9) times per hour, burrow inhabitants would thus normally be detected by the system within a given hour. The total number of readings, including all individuals (i.e., males and females) by the RFID-system was 1,846,831.

Defining local density and probability to attract opposite sex partners

We calculated each individual's attractiveness each hour (t) throughout the study period. We achieved this by scoring, for each hour, whether a female was detected in the burrow occupied by a focal male as a binary variable (0 = no, 1 = yes). For each burrow and hour, we also calculated the local density of males (see below) defined as the number of males (including the focal male) per m² detected by the entire system during the previous hour ($t-1$). We focused on local male density since males primarily compete acoustically for mating opportunities (Gerhardt and Huber 2002). Competitive regimes experienced by males therefore vary with male, not overall, density. In addition, females do not produce acoustic signals, implying that male crickets may not easily assess female density. Male crickets generally stop responding to acoustic signals of other males produced over 5 m away (Simmons 1988; Hissmann 1991), potentially limiting acoustic competition between males within this spatial scale. As we did not have prior information on the exact range over which male respond to local male density, we investigated which density radius best explained our data (see below) (see Wilkin et al. 2006 for a similar approach). We initially considered 1, 2, 3, 4, and 5 m as potential radii. However, since the model with a 5-m local density radius was the best fitting model (see Results), we fitted two post hoc models with larger radii (6 and 7 m). This enabled us to test whether the 5-m model really was truly best-fitting. We then selected the best-fitting model for analyses presented in the main text (see “Statistical methods”).

We collected population density data for the years 2014 and 2015. We have RFID data for the whole population only for the year 2015. In the year 2014, the number of adult crickets in our study plot was 265 individuals (i.e., 0.106 adults/m²) while in year 2015, it was 90 individuals (i.e., 0.036 adults/m²). In a nearby study area, an open study population initiated in 2018, the densities were 0.098 (for 2018), 0.053 (for 2019), and 0.077 (for 2020) (Niemelä, unpublished data). We thus considered the focal year of study (2015) to be a relatively “low-density” year (leading to our predictions: see Introduction).

There were 1, 7, 18, 18, 21, 26, and 31 focal burrows (out of 76 burrows in the final data set; see below) within the 1, 2, 3, 4, 5, 6, and 7 m radius, respectively, that had surface areas partly outside

(i.e., overlapping with the fenced border of) our study plot. For such burrows, we also performed our density calculations using only the surface area inside our study plot. We statistically controlled for the “edge” effect by including distance between the edge and the focal burrow as a covariate in all our models.

We only used data from individuals marked directly upon maturation and for which we could thus calculate adult age exactly. Age affects many traits in our model species, including the sexual signaling and attractiveness (Verburt et al. 2011; Rodríguez-Muñoz et al. 2019), and was therefore modeled. We consequently removed males from the data set that were found, and marked, as adults of unknown age ($n = 7$ individuals). Notably, those males with unknown exact adult age were included in local male density calculations. Our final sample size was 7603 (hourly) data points collected over a 30-day period from 24 males occupying 76 unique burrows fitted with antennae. Each male occupied on average 5.75 (SD = 2.74) unique burrows and changed burrow 10.03 (SD = 7.29) times during its adult life. The average male was sharing a burrow with on average 5.00 unique females (SD = 3.04; range = 0–10). When a female was detected at a focal burrow, she stayed on average for 16.38 h (SD = 33.91; range = 1–238 h). In other German field cricket populations, ~70% of the adult males live less than 20 days (Ritz and Köhler 2007). This means that our sampling period of 30 days (36 days for longevity data) covered the majority of the species’ adult lifespan.

Statistical methods

Defining local density and estimating density-dependent attractiveness

We first explored over which range (radius) local male density best explained variation in attractiveness. Importantly, for testing our main hypothesis, that is, we asked whether males that are attractive at low density are less attractive at high density and the other way around, we estimated individual level reaction norms describing the change in attractiveness as a function of local density of competitors by using random regression models (Schaeffer 2004). These random regressions estimated the among-individual variance in intercepts and slopes of the reaction norms (see below), but also the covariance between these reaction norm components (Dingemanse et al. 2010). We used these covariances to test for the predicted trade-off in attractiveness between low and high densities. Specifically, we a priori predicted a negative covariance between intercepts and linear slopes of the attractiveness-density reaction norm. In more detail, we considered that individuals could differ in three components of the function describing how their hourly attractiveness vary as a function of density (i.e., reaction norm): 1) reaction norm intercept (i.e., elevation of the reaction norm), 2) linear component of their reaction norm slope, and 3) quadratic, that is, nonlinear, component of their reaction norm slope. We considered quadratic reaction norm shapes because density-dependent effects are often nonlinear (Stephens et al. 1999). We ran one model for each density range. Finally, for each reaction norm component, we further extract the BLUPs (i.e., best linear unbiased predictors; Henderson 1975) from the best fitting model, which we used to estimate whether linear selection is acting on these components.

To select the spatial range under which the main hypothesis was tested, we ran seven separate univariate random regression mixed effects models that fitted hourly attractiveness as a binary response variable and local male density as a fixed effect covariate, the latter calculated using a 1, 2, 3, 4, 5, 6, and 7 m radii, respectively.

Additionally, we ran a (null) model not including any of the local density metrics. To control for potential confounding effects, all models included distance from the edge (one value for each burrow; centimeters), age (for each male for each day; days), squared age, maximum age (one value for each male; days), squared maximum age, male density (for each male for each hour; no. males/m²), squared male density, maximum density (one value for each male), and squared maximum density as continuous covariates. We included squared effects for age and local density since nonlinear effects are often predicted (Stephens et al. 1999; Rodríguez-Muñoz et al. 2019). Maximum (squared) age controls the estimates for selective disappearance of individuals (Bouwhuis et al. 2009). Fitting (squared) maximum density enabled us to avoid bias in estimates of density caused by individual variation in density (Bouwhuis et al. 2009; Dingemanse et al. 2020). We did not control for minimum density because this variable did not vary among individuals. We included individual identity (average number of data points per individual [SD]; 316.79 [124.36]), date (253.43 [89.31]), hour (316.79 [76.07]), and burrow identity (100.04 [102.63]) as random effects to estimate variation in response variables caused by individual and any (unmeasured) spatiotemporal environmental effects. Finally, to test our main hypothesis, we included the *interaction* (i.e., random slopes) between local male density (linear and quadratic terms) and individual identity as an additional random term. This interaction term allowed estimating individual variation in plasticity in attractiveness with respect to density, and covariances between all reaction norm components (i.e., intercepts, linear slopes, quadratic slopes).

Model selection was used to compare the relative fit of models that differed solely in how density was defined, using the Watanabe-Akaike’s Information Criterion (WAIC) as a measure of model fit (Gelman et al. 2014). Models with Δ WAIC > 7 were considered as fitting the data significantly worse than the best-fitting model (Gelman et al. 2014).

Estimating directional selection on attractiveness reaction norms

We extracted best linear unbiased predictors (BLUPs) for each individual’s attractiveness-density reaction norm intercept, linear slope, and quadratic slope, and subsequently tested their independent effects reproductive success (i.e., total number of unique female partners; see Introduction). Specifically, we tested independent effects of abovementioned reaction norm components on 1) total number of unique female partners and 2) total number of unique female partners while additionally controlling for longevity. If any relationship of reaction norm components with reproductive success resulted from some crickets living longer, any effect should disappear in model fitting longevity as an additional fixed effect (compared to model ii). We fitted the three reaction norm components as covariates into this linear multiple regression, fitting two separate models (i.e., models 1) & 2) described above). To acquire estimates of standardized linear (i.e., directional) selection gradients, the reproductive success (i.e., total amount of unique partners) was expressed in terms of relative fitness by dividing each value by the population-mean. Notably, an individual’s reaction norm components for density-dependent attractiveness are estimated at zero density, because density was naturally left-centered in our reaction norm analyses from which this parameter was extracted (see section *Defining local density and estimating density-dependent attractiveness*). For the selection gradient model, the fitted reaction norm BLUPs

were mean-centered and standardized prior to selection gradient analysis (Lande and Arnold 1983). These two multiple regressions thereby estimated direct linear (directional) selection gradients on each reaction norm component, not attributable to the correlations between the components.

The usage of BLUPs has been criticized because they are estimated with large error when based on few repeats per individual and this error is not taken forward to subsequent analyses (Houslay and Wilson 2017). The BLUPs extracted here were estimated based on an average sample size of 316.79 (SD = 124.36) measurements (i.e., hours) per individual, effectively making such problems irrelevant in our case. Moreover, simulations have shown that approaches where uncertainty in BLUPs, fitted as predictors, is taken forward to subsequent analyses might bias estimates (Dingemanse et al. 2020); we therefore did not attempt to control for error in estimated BLUP values. Notably, despite having acquired hundreds of measurements of each individual, our sample size in terms of individuals is rather low ($n = 24$ adult males). This may have resulted in increased type II error rates. Marginally nonsignificant patterns should thus be treated with caution, as we might have had insufficient statistical power to detect weak effects.

All models were run in the R (version 3.6.1.) statistical environment (R Core Team 2019). We fitted random slopes models in the package brms (version 2.10.0.) (Bürkner 2017) using a binomial link function. We used default, student-T, priors and the model was run with 3000 iterations with 1000 burn-in and sampling rate of 2, resulting in low autocorrelation between samples. We used the package lme4 (version 1.1.20.) (Bates et al. 2015) with Gaussian link functions to estimate selection gradients.

RESULTS

Density dependence of attractiveness at the population level

Of the seven alternative definitions of density, the model using a 5-m radius around the male's burrow fitted the data best (Table 1, Supplementary Table S1). The model not fitting local density had the worst fit (Table 1, Supplementary Table S1). We based all subsequent interpretations and subsequent analyses on this best-fitting model. Note that we do not focus on interpretation of all fixed effects (included in the model to avoid parameter bias); rather, we discuss the random effects structure as those are important for testing our hypotheses. Estimates for local densities for 1–4, 6, and 7 m radii and for the model without density are given in the Supplementary Material (Supplementary Table S1). It is worth

Table 1
WAIC values with standard error (SE: in brackets) for all models fitted to the data

Model	WAIC (SE)
1 m radius	4081.37 (97.44)
2 m radius	3969.86 (97.38)
3 m radius	3986.15 (95.88)
4 m radius	3916.18 (96.15)
5 m radius	3886.36 (97.63)
6 m radius	3903.72 (97.32)
7 m radius	3933.95 (94.86)
No density	4125.34 (97.40)

noting that all models gave qualitatively the same results for the key parameter estimates.

The best-fitting model showed significantly positive linear, and significantly negative nonlinear, effects of male age on hourly probability to attract a female (Table 2); males became more attractive with increasing age but this effect leveled off when ageing, confirming that age (or experience) is important factor contributing male attractiveness (Rodríguez-Muñoz et al. 2019). There was no selective disappearance of males from the data based on a lack of maximum age effect (Table 2).

Attractiveness (moderately) increased versus (steeply) decreased with linear versus quadratic effects of local density, respectively (Table 2). This means that attractiveness was greatest under intermediate densities. A plot of the point estimates confirms this interpretation (black line in Figure 2). These findings are overall consistent with the existence of an Allee effect, as attractiveness increased from low to intermediate local densities (Courchamp et al. 1999; Stephens et al. 1999). Attractiveness declined from intermediate to high density (Figure 2).

Individual variation in attractiveness-density reaction norms

There was individual variation in overall hourly attractiveness owing to among-individual variance in reaction norm intercept (Table 2). Males also differed in how their attractiveness changed with density. Specifically, both linear and nonlinear reaction norm slopes differed between individuals (Table 2; Figure 2a). As predicted by our main

Table 2

Sources of variation in attractiveness for the best fitting model (where density was defined over a 5-m radius); we present fixed (β) and random (σ^2) parameters with 95% credible intervals, derived from a univariate random slopes mixed effects model

Fixed effects*	5 m radius
	β (95% CI)
Intercept	−4.654 (−28.038, 7.114)
Distance from edge	−0.004 (−0.007, −0.001)
Age	0.533 (0.335, 0.737)
Age ²	−0.008 (−0.013, −0.003)
Maximum age	0.948 (−0.518, 2.821)
Maximum age ²	−0.017 (−0.054, 0.013)
Maximum density	−590.835 (−1225.166, 24.769)
Maximum density ²	5579.077 (−445.705, 11901.782)
Density	106.808 (32.623, 186.338)
Density ²	−1431.068 (−2152.257, −703.362)
Random effects	
Variance	σ^2 (95% CI)
Burrow	21.317 (13.001, 35.831)
Date	8.506 (4.522, 18.175)
Hour	0.001 (<0.001, 0.004)
Individual	21.855 (9.999, 50.606)
Individual linear slope*	13.368 (6.529, 29.070)
Individual nonlinear slope	3.966 (0.007, 118.552)
Correlations	
$r_{\text{intercept,slopes}}$ (95% CI)	
Intercept-linear slope	−0.588 (−0.827, −0.212)
Intercept-nonlinear slope	0.010 (−0.866, 0.880)
Linear slope-nonlinear slope	0.003 (−0.865, 0.888)

*Multiply by 1000.

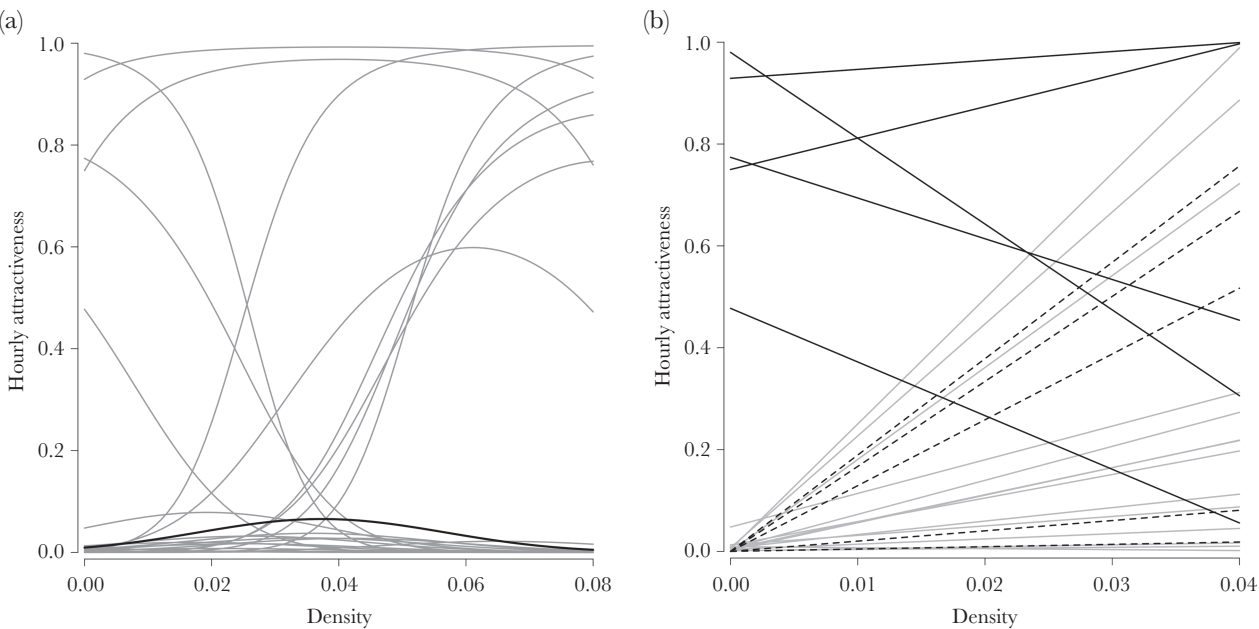


Figure 2 Model predictions of (a) individual-level reaction norms representing attractiveness as a function of local density and (b) individual-level reaction norms for a section of the data ranging from zero to intermediate density (fitting linear reaction norms only) to help interpret the negative correlation between reaction intercepts and linear slopes. The bold black line in (a) represents the population-average level of attractiveness across local densities. In (b), to help to visualize the negative intercept-linear slope correlation, black lines refer to 5 males with highest values for reaction norm intercept, black dashed lines refer to 5 males with lowest values for intercept, while the rest 14 gray lines represent all other males. Estimates for the reaction norms and population level regression line are taken from the best fitting random slopes mixed effects model (Table 2).

hypothesis, the intercept and linear slope of the reaction norm were negatively correlated (Table 2). This negative correlation does not necessarily imply support for our main prediction, positing that some males are most attractive at low density and other males are most attractive at high density. In fact, when transformed to their absolute values, only three males had negative linear slopes. For those males only, attractiveness decreased with increasing local density (Figure 2). A plot of model predictions indicated that an alternative interpretation of the negative correlation between intercept and linear slope fitted the data best: attractiveness generally increased with increasing density but less steeply for individuals that were already more attractive at the lowest densities. To illustrate this finding, we plot the linear reaction norms estimates over this limited density range in Figure 2b and Supplementary Figure S3. Importantly, since the local density was naturally left-centered, variation in all reaction norm components should be interpreted as being present at the zero-density.

Selection acting on attractiveness

The phenotypic selection analysis showed no statistical evidence for directional selection on any of the reaction norm components (Table 3 (a), Figure 3). That is, none of them predicted variation in the total number of unique partners acquired during adult life. Including longevity in the model also did not change this pattern (Table 3 (b)). However, longevity did positively predict the total amount of unique partners (Table 3 (b)), indicating that longevity contributed to male fitness as predicted.

DISCUSSION

We tested a key prediction of theory predicting density-dependent trade-offs in attractiveness. We expected that those males that are

Table 3 Sources of variation in (a) the total amount of potential mating partners and (b) the total amount of potential mating partners, while controlling for longevity, explained by an individual’s intercept, linear, and nonlinear component of density-dependent attractiveness reaction norms. We present fixed (β) parameters with standard errors, as well as test statistics (T -value) with accompanying P -values, derived from a general linear models. Fitness is transformed in relative fitness and traits centered and standardized. β thus represents the standardized selection gradient for each “trait” (i.e., reaction norm component)

(a)	β (SE)	T -value	P -value
Model intercept	1.00 (0.126)	7.916	>0.001
RN intercept	0.06 (0.189)	0.307	0.762
RN linear slope	−0.05 (0.193)	−0.249	0.806
RN quadratic slope	0.21 (0.146)	1.459	0.160
(b)	β (SE)	T -value	P -value
Model intercept	1.00 (0.111)	9.013	>0.001
RN intercept	0.06 (0.166)	0.363	0.720
RN linear slope	0.02 (0.171)	0.142	0.888
RN quadratic slope	0.13 (0.132)	1.020	0.321
Longevity	0.31 (0.117)	2.632	0.016

attractive at low densities would be less attractive at high densities, and that males that are attractive at high densities would be less attractive at low densities. We did not find support for this prediction. Instead, males generally increased their attractiveness with increasing density, at least from low to intermediate densities, revealing the presence of a population-level Allee effect. We did,

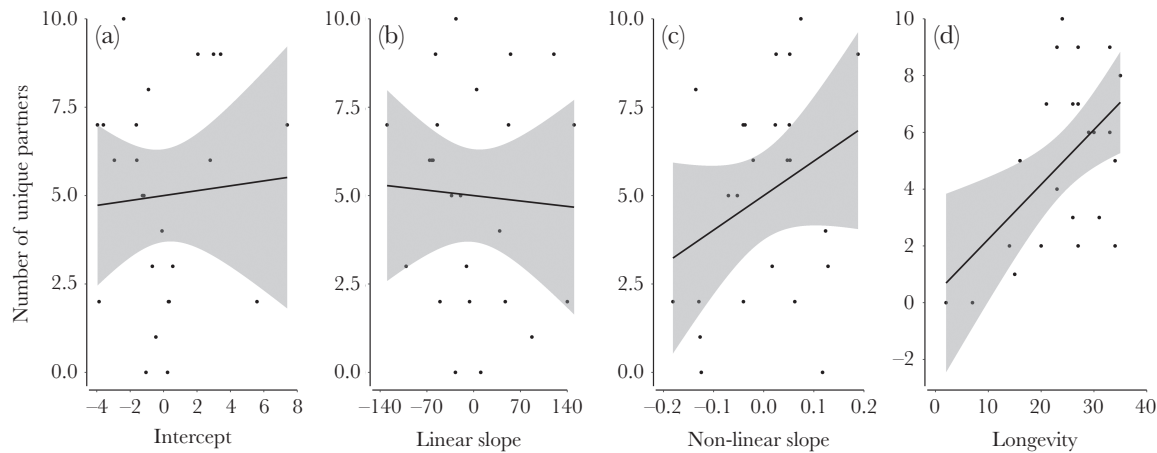


Figure 3

The number of unique partners plotted as a function of individual-level estimates (BLUP) for the (a) intercept, (b) linear, and (c) nonlinear slopes of attractiveness-density reaction norms, as well as (d) longevity (Table 3). For (a) to (c), the black dots are residuals from the models where each focal component of the reaction norm (BLUP) is fitted as a separate response variable from three separate models, and where the other two components are set as covariates. Any nonindependence between reaction norm components (BLUPs) was therefore controlled for. For (d), raw values for longevity are used. The solid black line is a regression line describing the association between y and x and the shaded gray area refers to standard errors.

nevertheless, find support for our prediction that density-dependent changes in attractiveness differed between males, however, the pattern was not of a form that we had expected. Specifically, males that were less attractive at the lowest densities showed a greater increase in attractiveness with density compared to more attractive males at the lowest density. Finally, we did not find evidence for directional selection acting on this variation based on the fitness metrics used in this study (i.e., number of unique partners). Overall, our study did not confirm the existence of the hypothesized trade-off, mediated by variation in local density, maintaining individual variation in behavior and life history.

Density dependence and the Allee effect at the population level

Our results show that attractiveness changed with density as predicted by the Allee effect theory because attractiveness increased from low to intermediate density (Allee 1931; Stephens et al. 1999). The most likely mechanism for the increasing attractiveness from low to intermediate densities is that generally larger acoustic choruses of males attract proportionally more females, increasing the probability of each male to attract a female (Gerhardt and Huber 2002; Isvaran and Pongshe 2013). Indeed, the number of visiting females has been shown to increase with increasing local density in chorusing species (Ryan et al. 1981, but see Gerhardt and Huber 2002). Our results also show that a male's ability to attract females generally declines at high densities. This pattern likely results from increasing effects of male–male acoustic competition; under high local densities females are not able to discriminate between males due to masking interference of acoustic signals, potentially leading to suboptimal mating decisions (Gerhardt and Huber 2002). Indeed, female choice might become more difficult at high densities due to cognitive limitations (Kokko and Rankin 2006). Thus, increased acoustic competition might also lead females to avoid extremely high male densities. Moreover, high male densities might also come with more physical (aggressive) interactions between males (c.f. Rodríguez-Muñoz et al. 2010), decreasing the time available for attracting mating partners. These positive (Allee effect) and negative (competition) effects combined likely explain the overall

concave shape of how attractiveness changed with local density at the population level.

Individual differences in density-dependent attractiveness

Male crickets differed in average attractiveness and in how their attractiveness changed with local competitor density, implying the existence of individual differences in mating strategies and/or ability to reap social benefits from the presence of other males. Crickets are known to use a diverse array of mating strategies (Cade 1981; Simmons et al. 2010; Pascoal et al. 2014). Individual attractiveness has not been linked to natural variation in population density before in this detail in a wild insect population. The mechanism underpinning an individual's density-dependent change in attractiveness likely relates to how a male changes its acoustic signaling with density. Acoustic signaling represents a key channel by which both male and female crickets acquire information about the structure of the social environment (Gerhardt and Huber 2002). Opposite to our main prediction, we did not find evidence for the existence alternative strategies (Lande et al. 2009; Réale et al. 2010; Engen et al. 2013; Dammhahn et al. 2018; Wright et al. 2019), at least with respect to the within-year longitudinal variation in competitive regimes studied here. This mismatch between the predictions generated by formal (and verbal) models and our results may relate to the variation in density in our populations across generations, where such trade-offs are realized or evolve, respectively. Indeed, in Engen et al.'s (2013) formal model, population densities showed 16-fold changes across generations: if such degrees of variation do not exist in our population, predicted density-dependent strategies might also not evolve. To date, there are only a handful of empirical papers, restricted to few taxonomic groups (e.g., birds) that have tested the predictions of these density-dependent models. General conclusions, therefore, cannot yet be drawn.

Interestingly, males that were less attractive at very low densities increased their attractiveness from low toward intermediate densities more steeply than males that were already attractive. In other words, the magnitude of the positive effect of local density on attractiveness (i.e., the Allee effect) varied

between individuals. This effect may be driven by differential increase in investment in acoustic signaling by males with low attractiveness at low density when the local density increases (see above). Interestingly, given that the among-individual variation in the Allee effect would be underpinned by additive genetic variation, the Allee effect should have the *potential* to evolve. This potential is likely there since half of the among-individual variance in behaviors is underpinned by additive genetic variation (Dochtermann et al. 2015). Translating evolutionary potential into evolution change would require selection acting on the types of reaction norms we studied in our crickets. This is a key area for future research. At the moment, individualized estimates of Allee effects are extremely rare in the literature probably because the collection of such data requires the estimation of individual variation in (nonlinear) reaction norms; doing so requires many repeated measurements per individual, across the entire environmental gradient (Nussey et al. 2007; Dingemanse et al. 2010; van de Pol 2012). With the increasing usage of automated data collection platforms, such work is becoming increasingly feasible, as demonstrated by this study.

Selection acting on density dependence of attractiveness

We found that reproductive fitness, here measured using the total amount of unique female partners observed with each focal male (Rodríguez-Muñoz et al. 2010) did not differ statistically between males with different reaction norms: none of the reaction norm components were under directional selection. The sample size of this analysis was small, however, since our point estimates for selection gradients were close to zero, we believe that this result does not represent a Type II error. Perhaps a more fitting explanation should center on the notion that annual variation in fitness related to these strategies does not fully capture natural selection, implying that longitudinal among-year datasets are required. It is, for example, conceivable that the year of study was characterized by a local density within a range that does not induce selection on the behavioral reaction norm under study here. Indeed, other studies imply that patterns of (e.g., frequency-dependent) selection are strongest when competition for resources is intense rather than relaxed (Sinervo et al. 2000). Interestingly, our finding of equal fitness of males with varying density-dependent attractiveness can itself maintain variation in attractiveness-density reaction norms.

We note that males that lived longer were also seen with more unique partners over their entire lifetime. Longevity is generally considered an important determinant of lifetime-reproductive success because individuals living long might be of higher quality, and have more opportunities to meet many mating partners (Reid et al. 2003; Rodríguez-Muñoz et al. 2019). Future work in this area may focus on four main avenues. First, are reaction norms of social behaviors heritable? Second, do patterns of selection on social reaction norms vary as a function of both within- and among-year changes in density? Third, which phenotypic traits underpin the observed individual differences in attractiveness reaction norms? Finally, are these patterns of phenotypic integration adaptive and maintained by natural selection?

Study limitations

One of the key limitations of our study is that that we presented data for only a single year. We fully appreciate that study

replication (over multiple years or sites) would be formally required to firmly demonstrate density-dependent selection on attractiveness reaction norms and consider this element as an outlook for future research in this area. The expression of trade-offs might also depend on the overall population density, varying across years (see Methods), which might why we did not detect them. Firm studies demonstrating such trade-offs may require an experiment manipulating male density is manipulated and attractiveness under a range of density.

Observational work conducted in the wild faces many challenges when it comes to data interpretation. First labile, repeatedly expressed behaviors collected in the wild might be biased by plethora of confounding environmental effects (Niemelä and Dingemanse 2017, Hertel et al. 2020). The potential for such biases can be reduced by including spatiotemporal variation in statistical models. For this very reason, we fitted burrow, date and hour as random effects in our models (Niemelä and Dingemanse 2017; Hertel et al. 2020). Secondly, interpretation of the parameter estimates from observational data is more difficult compared to those derived from manipulated experiments because effects of confounding predictors cannot be separated. For example, while increased acoustic competition at high densities might be the most likely explanation for the low attractiveness at high densities at the population level in our model species (Gerhardt and Huber 2002; Rodríguez-Muñoz et al. 2010), density-dependent female choice can also lead to density-dependent variation in attractiveness (Kokko and Rankin 2006). In fact, under some scenarios, both density-dependent male-male competition and density-dependent female choice can lead to population-level attractiveness reaction norms of similar shapes (Kokko and Rankin 2006), making their respective effects on attractiveness difficult to separate in observational data. The issues highlighted above imply that care is required when designing and interpreting observational studies conducted in the wild.

CONCLUSION

Our longitudinal study of a natural insect population did not support our main prediction concerning the existence of hypothesized density-dependent strategies. Reaction norm analyses did nevertheless provide evidence for individual variation in density dependence of attractiveness, highlighting the importance of considering socioecological context. Importantly, our work confirms that it is possible to use state-of-the-art surveillance technology to track small insects in their natural habitats to collect detailed spatiotemporal information about their social interactions throughout their lives (see also Molet et al. 2008; Rodríguez-Muñoz et al. 2011; Tenczar et al. 2014). Insects can help to seek answers to many complex evolutionary and ecological questions, such as those related to social selection and evolution.

SUPPLEMENTARY MATERIAL

Supplementary data are available at *Behavioral Ecology* online.

FUNDING

P.T.N. and N.J.D. were supported by the DFG (Deutsche Forschungsgemeinschaft; grant numbers NI 1539/2-1 and DI1694/1-1). S.T. was supported by the LMU (Ludwig-Maximilians University of Munich).

Author contributions: P.T.N. and N.J.D. contributed to the design of the study and the drafting and revising of the article. P.T.N. collected and analyzed all the data. P.T.N. and S.T. contributed to the extraction of the data. All authors approved the final version of the manuscript.

Conflict of interest: Authors declare no conflict of interest. We thank the Editor and three anonymous reviewers for constructive feedback on the manuscript.

Data availability: The data, code for statistical models and the model objects are available in the The Open Science Framework, that is, OSF (<https://osf.io/ytrpf9/>). Data are also available in DRYAD: Analyses reported in this article can also be reproduced using the data provided by (Niemelä et al. 2020). We confirm that, for all experiments, all measures, conditions and data exclusions are reported.

Handling editor: Emilie Snell-Rood

REFERENCES

- Allee WC. 1931. Animal aggregations, a study in general sociology. Chicago (IL): Univ. of Chicago Press.
- Andersson M. 1994. Sexual selection. Princeton (NJ): Princeton University Press.
- Bates D, Maechler M, Bolker B, Walker S. 2015. Fitting linear mixed-effects models using lme4. *J Stat Softw*. 67:1–48.
- Bateson M, Healy SD. 2005. Comparative evaluation and its implications for mate choice. *Trends Ecol Evol*. 20:659–664.
- Bouwhuys S, Sheldon BC, Verhulst S, Charmantier A. 2009. Great tits growing old: selective disappearance and the partitioning of senescence to stages within the breeding cycle. *Proc Biol Sci*. 276:2769–2777.
- Bürkner P. 2017. brms: an R package for Bayesian multilevel models using stan. *J Stat Softw*. 80:1–28.
- Cade WH. 1981. Alternative male strategies: genetic differences in crickets. *Science*. 212:563–564.
- Courchamp F, Clutton-Brock T, Grenfell B. 1999. Inverse density dependence and the Allee effect. *Trends Ecol Evol*. 14:405–410.
- Dammhahn M, Dingemanse NJ, Niemelä PT, Réale D. 2018. Pace-of-life syndromes: a framework for the adaptive integration of behaviour, physiology and life history. *Behav Ecol Sociobiol*. 72:62.
- Dingemanse NJ, Kazem AJN, Réale D, Wright J. 2010. Behavioural reaction norms: animal personality meets individual plasticity. *Trends Ecol Evol*. 25:81–89.
- Dingemanse NJ, Moiron M, Araya-Ajoy YG, Mouchet A, Abbey-Lee RN. 2020. Individual variation in age-dependent reproduction: fast explorers live fast but senesce young? *J Anim Ecol*. 89:601–613.
- Dochtermann NA, Schwab T, Sih A. 2015. The contribution of additive genetic variation to personality variation: heritability of personality. *Proc R Soc B*. 282:20142201.
- Duckworth RA, Badyaev AV. 2007. Coupling of dispersal and aggression facilitates the rapid range expansion of a passerine bird. *Proc Natl Acad Sci USA*. 104:15017–15022.
- Engen S, Lande R, Saether BE. 2013. A quantitative genetic model of r- and K-selection in a fluctuating population. *Am Nat*. 181:725–736.
- Fisher DN, David M, Tregenza T, Rodríguez-Muñoz R. 2015. Dynamics of among-individual behavioral variation over adult lifespan in a wild insect. *Behav Ecol*. 26:975–985.
- Fisher DN, James A, Rodríguez-Muñoz R, Tregenza T. 2015. Behaviour in captivity predicts some aspects of natural behaviour, but not others, in a wild cricket population. *Proc Biol Sci*. 282:20150708.
- Gawalek M, Dudek K, Ekner-Grzyb A, Kwieciński Z, Sliwowska JH. 2014. Ecology of the field cricket (Gryllidae: Orthoptera) in farmland: the importance of livestock grazing. *North West J Zool*. 10:325–332.
- Gelman A, Hwang J, Vehtari A. 2014. Understanding predictive information criteria for Bayesian models. *Stat Comput*. 24:997–1016.
- Gerhardt CH, Huber F. 2002. Acoustic communication in insects and anurans. Chicago (IL): The University of Chicago Press.
- Henderson CR. 1975. Best linear unbiased estimation and prediction under a selection model. *Biometrics*. 31:423–447.
- Hertel AG, Niemelä PT, Dingemanse NJ, Mueller T. 2020. A guide for studying among-individual behavioral variation from movement data in the wild. *Mov Ecol*. 8:1–18.
- Hissmann K. 1991. Phonotaxis of male crickets (*Gryllus campestris*) in a field population as an indication of territoriality (Orthoptera: Gryllidae). *J Insect Behav*. 4:675–681.
- Houslay TM, Wilson AJ. 2017. Avoiding the misuse of BLUP in behavioural ecology. *Behav Ecol*. 28:948–952.
- Isvaran K, Ponskhe A. 2013. How general is a female mating preference for clustered males in lekking species? A meta-analysis. *Anim Behav*. 86:417–425.
- Kokko H, Rankin DJ. 2006. Lonely hearts or sex in the city? Density-dependent effects in mating systems. *Philos Trans R Soc B*. 361:319–334.
- Lande R, Arnold SJ. 1983. The measurement of selection on correlated characters. *Evolution*. 37:1210–1226.
- Lande R, Engen S, Saether BE. 2009. An evolutionary maximum principle for density-dependent population dynamics in a fluctuating environment. *Philos Trans R Soc Lond B Biol Sci*. 364:1511–1518.
- Molet M, Chittka L, Stelzer RJ, Streit S, Raine NE. 2008. Colony nutritional status modulates worker responses to foraging recruitment pheromone in the bumblebee *Bombus terrestris*. *Behav Ecol Sociobiol*. 62:1919–1926.
- Nicolaus M, Tinbergen JM, Ubels R, Both C, Dingemanse NJ. 2016. Density fluctuations represent a key process maintaining personality variation in a wild passerine bird. *Ecol Lett*. 19:478–486.
- Niemelä PT, Dingemanse NJ. 2017. Individual versus pseudo-repeatability in behaviour: lessons from translocation experiments in a wild insect. *J Anim Ecol*. 86:1033–1043.
- Niemelä PT, Lattenkamp EZ, Dingemanse NJ. 2015. Personality-related survival and sampling bias in wild cricket nymphs. *Behav Ecol*. 26:936–946.
- Niemelä PT, Tiso S, Dingemanse NJ. 2020. Density-dependent individual variation in male attractiveness in a wild field cricket. *Behav Ecol*. doi:10.5061/dryad.83bk3j9qq.
- Nussey DH, Wilson AJ, Brommer JE. 2007. The evolutionary ecology of individual phenotypic plasticity in wild populations. *J Evol Biol*. 20:831–844.
- Pascoal S, Cezaud T, Eik-Nes A, Gharbi K, Majewska J, Payne E, Ritchie MG, Zuk M, Bailey NW. 2014. Rapid convergent evolution in wild crickets. *Curr Biol*. 24:1369–1374.
- van de Pol M. 2012. Quantifying individual variation in reaction norms: how study design affects the accuracy, precision and power of random regression models. *Methods Ecol Evol*. 3:268–280.
- R Core Team. 2019. R: a language and environment for statistical computing. Vienna (Austria): R Found Stat Comput.
- Réale D, Garant D, Humphries MM, Bergeron P, Careau V, Montiglio PO. 2010. Personality and the emergence of the pace-of-life syndrome concept at the population level. *Philos Trans R Soc Lond B Biol Sci*. 365:4051–4063.
- Reid JM, Bignal EM, Bignal S, McCracken DI, Monaghan P. 2003. Age-specific reproductive performance in red-billed choughs *Pyrrhocorax pyrrhocorax*: patterns and processes in a natural population. *J Anim Ecol*. 72:765–776.
- Ritz MS, Köhler G. 2007. Male behaviour over the season in a wild population of the field cricket *Gryllus campestris* L. *Ecol Entomol*. 32:384–392.
- Rodríguez-Muñoz R, Boonekamp JJ, Liu XP, Skicko I, Fisher DN, Hopwood P, Tregenza T. 2019. Testing the effect of early-life reproductive effort on age-related decline in a wild insect. *Evolution*. 73:317–328.
- Rodríguez-Muñoz R, Bretman A, Slate J, Walling CA, Tregenza T. 2010. Natural and sexual selection in a wild insect population. *Science*. 328:1269–1272.
- Rodríguez-Muñoz R, Bretman A, Tregenza T. 2011. Guarding males protect females from predation in a wild insect. *Curr Biol*. 21:1716–1719.
- Ryan MJ, Tuttle MD, Taft LK. 1981. The costs and benefits of frog chorusing behavior. *Behav Ecol Sociobiol*. 8:273–278.
- Sæther BE, Visser ME, Grøtan V, Engen S. 2016. Evidence for r- and K-selection in a wild bird population: a reciprocal link between ecology and evolution. *Proc R Soc B Biol Sci*. 283:20152411.
- Schaeffer LR. 2004. Application of random regression models in animal breeding. *Livest Prod Sci*. 86:35–45.
- Simmons LW. 1988. The calling song of the field cricket, *Gryllus bimaculatus* (de geer): constraints on transmission and its role in intermale competition and female choice. *Anim Behav*. 36:380–394.
- Simmons LW, Tinghitella RM, Zuk M. 2010. Quantitative genetic variation in courtship song and its covariation with immune function and sperm quality in the field cricket *Teleogryllus oceanicus*. *Behav Ecol*. 21:1330–1336.

- Sinervo B, Svensson E, Comendant T. 2000. Density cycles and an off-spring quantity and quality game driven by natural selection. *Nature*. 406:985–988.
- Stephens PA, Sutherland WJ, Freckleton RP. 1999. What is the Allee effect? *Oikos*. 87:185.
- Tenczar P, Lutz CC, Rao VD, Goldenfeld N, Robinson GE. 2014. Automated monitoring reveals extreme interindividual variation and plasticity in honeybee foraging activity levels. *Anim Behav*. 95:41–48.
- Verburgt L, Ferreira M, Ferguson JWH. 2011. Male field cricket song reflects age, allowing females to prefer young males. *Anim Behav*. 81:19–29.
- Wilkin TA, Garant D, Gosler AG, Sheldon BC. 2006. Density effects on life-history traits in a wild population of the great tit *Parus major*: analyses of long-term data with GIS techniques. *J Anim Ecol*. 75:604–615.
- Wright J, Bolstad GH, Araya-Ajoy YG, Dingemanse NJ. 2019. Life-history evolution under fluctuating density-dependent selection and the adaptive alignment of pace-of-life syndromes. *Biol Rev*. 94:230–247.